

Semiparametric Value-at-Risk and Expected Shortfall

with a Real-Time Misspecification Score

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Abstract

Value-at-Risk and Expected Shortfall, the conditional tail-risk forecasts at the center of market-risk management and its regulation, are commonly built from a fitted parametric return distribution, a benchmark that is hard to beat on most assets on most days. We develop a semi-parametric alternative that keeps the parametric conditional scale and replaces only its fixed innovation law. The replacement is a residual-quantile model, conditioned on real-time signals of residual-shape stress and amortized across assets, spliced to an extreme-value tail, with an optional conformal overlay for coverage. Amortization lets one fit serve hundreds of names and price newly listed assets on day one. The method's distinctive object is a misspecification frontier: a live, per-asset, per-day score that predicts when the flexible model beats the parametric standard. Its advantage is concentrated, not uniform. It is large in the high-stress states the score identifies, and there it replicates out of sample on an untouched 2000–2013 window and across foreign-exchange, country-index, and cross-asset universes. On the roughly ninety percent of calm asset-days it shrinks to near zero, and the flexible model matches the benchmark at no penalty. On accuracy its EVT-tailed layer attains the lowest joint (VaR, ES) loss against

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the standard daily benchmarks GARCH- t and FHS, a ranking that survives annual refitting. Exception counts are near nominal at both regulatory levels through the 2008 crisis. The score’s kurtosis component, fixed in advance, reproduces the concentration out of sample. The fuller composite we deploy is tuned in sample, and we read its holdout behavior as corroboration rather than an independent test.

Keywords: Value-at-Risk; Expected Shortfall; misspecification; quantile regression; extreme value theory; amortized estimation.

JEL classification: C14; C22; C52; C58; G17; G28.

Introduction

Value-at-Risk (VaR) and Expected Shortfall (ES) summarize the downside risk of a trading book: the lower-tail quantile of the conditional return distribution, and the average loss beyond it. Two research traditions produce these forecasts, and they impose different restrictions on the conditional return distribution. Econometrics builds parametric filters, such as GARCH and its leverage and skew extensions, whose conditional scale dynamics are excellent but whose innovation law is fixed by assumption. Machine learning builds distribution-free conditional quantile estimators, such as gradient-boosted quantile trees, neural implicit quantile networks (IQN), and generative Bayesian computation (GBC) (Polson and Sokolov, 2023), which assume nothing about the innovation law but must learn it from data. Learners of this kind now forecast conditional volatility better than fixed-form parametric models on average across large cross-sections (Zhang et al., 2024). Whether that average advantage reaches the regulatory tail, and on which days, is the question this paper takes up. Industry practice sits between the two. Under the Basel Fundamental Review of the Trading Book (FRTB), the workhorse internal models are historical simulation (HS) and its GARCH-filtered variant (FHS), and forecasts are judged by VaR exception counts at 99%

and 97.5%. The reported measure is Expected Shortfall at the 97.5% level. That figure is not a diagnostic. The reported ES sets the market-risk capital a bank holds against its book (Basel Committee on Banking Supervision, 2019), so errors are costly both ways. A forecast that runs high freezes capital that could fund positions, and one that runs low lifts the multiplier on the charge as exceptions accumulate. A method that claims to beat the industry standard must therefore beat HS and FHS on $ES_{97.5}$ under exception-test discipline. Beating a vanilla GARCH on average loss is not enough.

The advantage of a flexible model over this parametric benchmark is not spread evenly across days, and that unevenness organizes the paper. On the roughly ninety percent of asset-days (one name on one trading date) when residual-shape stress is low, the estimator we deploy and the parametric benchmark are statistically indistinguishable on forecast accuracy. Holding the flexible model carries no detectable penalty. Section 4 reports the intervals. The gains are concentrated in the minority of days when residual-shape stress is high. Locating that minority in advance, from an observable signal, is the paper’s central task. We construct such a signal, a real-time misspecification score, and show that it orders the flexible model’s advantage. We then build a risk model that exploits the ordering while matching the benchmark elsewhere. The generative members of the family go further: they deliver forecasts for assets that have no return history at all, which no parametric point-forecasting model can produce.

Recent-residual tail diagnostics, the excess kurtosis and asymmetry of the GARCH-standardized residuals, predict when distribution-free quantile methods improve on parametric VaR/ES. Where residual-shape stress is high, the flexible estimator has a large and significant advantage. Where it is low, the advantage is statistically indistinguishable from zero in the design panel and a small positive in the point-in-time universe, with no detectable loss. The score’s content is the ordering of magnitudes, with the edge concentrated in the top decile, not a hard threshold.

The paper contributes to three literatures. To the evaluation of VaR and ES forecasts and the comparison of competing conditional models, it offers the misspecification frontier: a real-time, per-asset, per-day score that predicts when distribution-free quantiles beat a GARCH- t benchmark, with per-date Diebold–Mariano significance. The frontier is validated across four universes: CRSP US equities (top-decile edge +2.98%, DM 10.5), foreign exchange (hyperinflation tier, pooled DM 4.12), twenty-six country indices (a developed-to-frontier-market gradient), and forty-three cross-asset instruments. The Korea-2026 crash serves as a falsification case that sharpens the claim, since price crashes are not residual misspecification. To global, pooled forecasting and cross-asset transfer, it shows that a single amortized shape model trained across hundreds of names carries transferable information about conditional tail shape. One fit replaces per-asset estimation of the innovation shape and transfers to names never seen in training. In its characteristics-only form it prices assets with no return history at all, the day-one regime in which no per-asset model exists. To semiparametric conditional tail-risk forecasting, it develops the estimator that captures the frontier. The design keeps the parametric filter’s scale, which is hard to improve, and replaces only its fixed innovation law where it fails. The forecast is a GARCH conditional scale times a state-conditioned nonparametric residual quantile, with an extreme-value left tail and an optional split-conformal overlay. The resulting estimator is co-best with CAViaR (Engle and Manganelli, 2004) in the 90% Model Confidence Set (Hansen et al., 2011) and improves significantly on GARCH- t , FHS, EWMA, HS, and GJR-skew- t . Its EVT-tailed accuracy layer attains the lowest joint (VaR, ES) FZ0 score against the standard daily benchmarks GARCH- t and FHS at both FRTB levels, a ranking that survives annual refitting, with exception counts near nominal through a 2008-crisis window. Two formal results motivate the frontier. A pinball regret identity shows that a forecaster’s excess loss is locally quadratic in its quantile wedge, so a wedge at a level generates regret there, scaled by the local density. A tail-wedge proposition shows that innovation-shape misspecification opens that

wedge in the tail.

The loss profile is asymmetric in the score. When the frontier is quiet the estimator ties or edges the parametric benchmark (calm-quintile edge ≈ 0). The top-decile improvement is +3.0%. On the hyperinflation currencies, where the parametric assumption breaks entirely, it is 12–14%. These are reductions in a proper scoring rule, not a monetary welfare measure. A percentage pinball edge is a percentage reduction in average quantile loss at the quoted level. The joint FZ0 score of Section 5 carries the economic comparison.

The paper proceeds as follows. Section 1 places the method against the industry benchmark and the forecasting literature. Section 2 develops the semiparametric estimator and the two results that motivate the frontier. Section 3 sets out the data and the forecast-evaluation protocol. Section 4 presents the misspecification frontier, and Section 5 reports the joint VaR and ES evaluation. Section 6 covers cross-sectional transfer, longer horizons, and cross-asset evidence, and Section 7 concludes.

1 Related literature and benchmark models

1.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t , the level- α return-space quantile is $q_t^\alpha = F_t^{-1}(\alpha)$ (negative in the left tail). The return-space Expected Shortfall is $e_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq q_t^\alpha]$ for continuous F_t . In general $e_t^\alpha = \alpha^{-1} \int_0^\alpha F_t^{-1}(u) du$, the coherent form used throughout. The regulatory VaR and ES are their negatives (the sign convention is fixed once in Section 2.1). FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at $\text{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket), while backtesting remains VaR-exception based. More than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a desk off the internal-models

approach. The empirical bar for “beating the industry standard” is therefore joint: accuracy on ES_{97.5} and pinball, together with unconditional (Kupiec, 1995) and independence (Christoffersen, 1998) exception tests at both levels. Significance is assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011).

1.2 The parametric family, the simulation family, and CAViaR

Our battery spans the models banks run. It includes historical simulation, EWMA/RiskMetrics (Zumbach, 2007), GARCH(1,1)- t (Bollerslev, 1986), and GJR-GARCH-skew- t (Glosten et al., 1993), which adds leverage and skewness. It also includes GARCH-FHS (Barone-Adesi et al., 1999), which pairs a parametric scale with empirical standardized residuals and is the volatile-regime best practice, and SAV-CAViaR (Engle and Manganelli, 2004), the strongest non-nested semiparametric rival among the classical benchmarks. The SAV (symmetric absolute value) specification models the quantile directly on the pinball loss, $q_t(\tau) = \beta_0 + \beta_1 q_{t-1}(\tau) + \beta_2 |r_{t-1}|$, the semiparametric analogue of a GARCH recursion written on the quantile.

1.3 Distribution-free conditional quantiles and GBC

The closest antecedent in this journal’s pages is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names. That transfer is evidence of a shared volatility mechanism, which our amortized design also exploits. The differences delimit this paper’s contribution. Their object is average realized-variance forecast performance. Ours is state dependence in conditional tail-risk forecast performance, with the full return distribution and its regulatory tail held to calibration, exception-test, and capital discipline. Where a flexible model helps is itself the question: on roughly ninety percent of asset-days it does not improve on the parametric benchmark, which cautions against extrapolating

average machine-learning superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at each listing age. On tabular state, gradient-boosted trees outperform the implicit quantile network, which we attribute to the different target and feature set.

The nonparametric family estimates $Q_\theta(\tau \mid x_t)$ directly by pinball-loss empirical risk minimization (ERM) (Koenker and Bassett, 1978). Its two members here are gradient-boosted quantile trees (GBM) and the implicit quantile network of Dabney et al. (2018), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). In the IQN the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time. Casting GBC as ERM conditional-quantile regression on observed pairs removes its forward-simulator requirement and extends it to observational time series. The deployed shape model is the GBM. The IQN is retained as a benchmark. Its generative reading (draw $\tau \sim U(0,1)$, output $Q_\theta(\tau \mid x)$) is a capability we note without using, since sampling and posterior uncertainty are outside this paper’s scope. Generalized Bayes enters through the Gibbs posterior $\pi_n(\theta) \propto \exp\{-\omega \sum_t \ell_\theta(z_t)\} \pi(\theta)$, which is built from a loss in place of a likelihood and targets the risk minimizer directly (Jiang and Tanner, 2008; Bissiri et al., 2016). Distribution-free finite-sample guarantees come from split-conformal and adaptive conformal inference (Gibbs and Candès, 2021), which recalibrate predicted intervals on held-out data so that coverage holds under exchangeability without a parametric distributional model.

2 Semiparametric VaR and ES model

Each estimator below can be reimplemented from the formulas and algorithm boxes given. A glossary of acronyms is in the Online Appendix.

2.1 Notation and objects

Let r_t be the return of an asset on day t and $\mathcal{F}_{t-1}$ the information available before t . The conditional τ -quantile of r_t given $\mathcal{F}_{t-1}$ is $Q_t(\tau) = \inf\{q : \Pr(r_t \leq q \mid \mathcal{F}_{t-1}) \geq \tau\}$, the number the return falls below with probability τ . One sign convention is used throughout. The paper works in *return space*, with $q_\tau = Q_t(\tau) < 0$ in the left tail and $e_\tau = \tau^{-1} \int_0^\tau Q_t(u) du \leq q_\tau$. This is the convention of the FZ scoring function ($v, e < 0$) and of each table, where ES values such as -6.37 are return-space conditional means. The regulatory *loss-space* numbers are their negatives, $\text{VaR}_\tau^{\text{loss}} = -q_\tau$ and $\text{ES}_\tau^{\text{loss}} = -e_\tau$, used only when quoting FRTB levels. The two are never mixed in one expression.

The *standardized residual* $z_t = (r_t - \mu_t)/\sigma_t$ is the return with the model’s conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law. The state vector s_t collects trailing realized volatility, the misspecification scores of Section 2.6, and lags, and is what the nonparametric quantiles condition on. A quantile level $\tau \sim \lambda$ is drawn from a base distribution λ . The learned generator $H_\phi(s, \tau)$ maps (state, level) to a quantile in the sense of Polson and Sokolov (2023): $z \stackrel{D}{=} H(s, \tau)$ with $\tau \sim U(0, 1)$ is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\}), \quad u = z - q, \tag{1}$$

whose expectation is minimized by a conditional τ -quantile (Koenker and Bassett, 1978)—uniquely when that quantile is unique—the way squared error elicits the mean. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once (the integrated loss is the continuous ranked probability score up to a factor of two, as recorded below). The loss is density-free, so each estimator below dispenses with a likelihood.

Choice of scoring rule. We score by pinball at the decision-relevant levels rather than by a global proper score, for three reasons. (a) The continuous ranked probability score (CRPS) is exactly the integral of the pinball loss over all levels, $\text{CRPS}(F, y) = 2 \int_0^1 \rho_\tau(y - F^{-1}(\tau)) d\tau$ (Gneiting and Raftery, 2007). Scoring by CRPS therefore weights quantile levels uniformly, so a 1% tail occupies 1% of the integral. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail, and CRPS would average those differences away against a body where all competitors agree. For the same reason, training uses tail-weighted τ -sampling. (b) Expected Shortfall is not elicitable on its own (no loss function is minimized by the true ES alone), but the pair (VaR, ES) is, under the joint scoring functions of Fissler and Ziegel (2016). The FZ-scored re-evaluation of the battery is an extension specified in advance (Section 5). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them.

2.2 The final model: the residual-hybrid estimator

Figure 1 summarizes the estimator’s stages and data flow. The paragraphs below define each stage.

The deployable model is a four-stage composition. The parametric filter estimates conditional scale, which it does well and with few parameters, and the nonparametric stage estimates the state-dependent shape of the innovation law, which the parametric family fixes by assumption. The extreme-value tail extrapolates beyond the estimation range using the generalized Pareto approximation of Section 2. The conformal stage corrects the level of the quantile curve using its measured miss rate on a held-out split, with the exchangeability-based guarantee of Proposition 1.

Motivation for the design. Conditional scale is identified by a few parameters and estimated with low variance by a parametric filter, so the shape stage is where flexible methods can add value. Per-asset tail data is scarce, which motivates pooling the standardized residuals across names. The EVT

stage extrapolates the tail by the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), and the conformal stage is motivated by the split-conformal guarantee of Proposition 1, exact under exchangeability and approximate for the deployed time-ordered residuals. These are the statistical reasons for each stage. The empirical sections test whether the combination delivers.

Stage 1 (parametric scale). Fit GARCH- t ,

$$r_t = \mu + \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2, \quad z_t \sim t_\nu \text{ (standardized)}, \quad (2)$$

by quasi-maximum likelihood on a trailing window, and keep $\hat{\mu}$, $\hat{\sigma}_t$, and the residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\hat{\phi} = \arg \min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau \sim \lambda} \rho_{\tau}(\hat{z}_{j,t} - q_{\phi}(\tau \mid s_{j,t})). \quad (3)$$

Stage 3 (EVT tail). Beyond the estimation range of (3), extrapolate with the generalized Pareto approximation that extreme-value theory motivates for exceedances over a high threshold, under the standard max-domain-of-attraction conditions (Pickands, 1975; McNeil and Frey, 2000). The ES formula below requires $\xi < 1$, a condition each fitted tail in this paper satisfies by a wide margin and the code guards explicitly.¹ On losses $X = -\hat{z}$, fix a threshold u at the empirical $(1 - p_0)$ loss quantile with $p_0 = 0.025$. Fit the generalized Pareto distribution (GPD) $G_{\xi, \beta}(x) = 1 - (1 + \xi x / \beta)^{-1/\xi}$ to exceedances $X - u > 0$ strictly on past data. The shape ξ sets how heavy the tail is and the scale β how wide. For $\tau \leq p_0$, set

$$\hat{q}^{\text{EVT}}(\tau) = -\left[u + \frac{\beta}{\xi}((\tau/p_0)^{-\xi} - 1)\right], \quad (4)$$

which is continuous at its own endpoint, $\widehat{q}^{\text{EVT}}(p_0) = -u$. The formula requires $\beta > 0$ and $1 + \xi(x - u)/\beta > 0$, and reduces to $-[u + \beta \log(p_0/\tau)]$ in the $\xi \rightarrow 0$ exponential limit. The threshold u is the *unconditional* empirical loss threshold, while the Stage-2 body quantile at p_0 is the state-conditional $\widetilde{q}(p_0 \mid s)$. The splice is therefore exactly continuous only when the threshold is anchored at the body's own p_0 -quantile, $u = -\widetilde{q}(p_0 \mid s)$. Otherwise a small level shift of size $|-u - \widetilde{q}(p_0 \mid s)|$ remains at the join, absorbed by the min-envelope and rearrangement in (5). The tail is applied selectively, where the raw tail under-covers, since it hurts some volatility-index series.

Stage 4 (conformal finish). On a held-out calibration split of size n , compute the signed errors $e_i = \widehat{z}_i - \widehat{q}(\tau \mid s_i)$. Let c_τ be the $\lceil (n+1)\tau \rceil$ -th order statistic of $\{e_i\}$. The curve is shifted per level by this empirical correction, $\widetilde{q}(\tau \mid s) = \widehat{q}(\tau \mid s) + c_\tau$ (Vovk et al., 2005; Romano et al., 2019). This distribution-free layer repairs the 97.5% level that each model in the battery otherwise fails.

Proposition 1 (split-conformal validity under exchangeability). *If the calibration and test residuals $e_1, \dots, e_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n+1)\tau \rceil \leq n$ the shifted quantile satisfies*

$$\tau \leq \Pr(z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})) = \frac{\lceil (n+1)\tau \rceil}{n+1} < \tau + \frac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of e_{n+1} among $\{e_1, \dots, e_{n+1}\}$ is uniform on $\{1, \dots, n+1\}$. The event $z_{n+1} \leq \widetilde{q}(\tau \mid s_{n+1})$ equals $\{e_{n+1} \leq e_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \leq \lceil (n+1)\tau \rceil\}$, whose probability is $\lceil (n+1)\tau \rceil / (n+1) \in [\tau, \tau + \frac{1}{n+1}]$ (Vovk et al., 2005). The full argument is in the Online Appendix, together with the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1) and the boundary convention $c_\tau = +\infty$ when $\lceil (n+1)\tau \rceil = n+1$. The guarantee is marginal, not conditional on s_{n+1} . $\square$

The proposition's scope is narrower than its use. In deployment the calibration split necessarily

precedes the test period, and time-ordered residuals are not exchangeable. The proposition therefore motivates the construction. The evidence that it works here is empirical: in the exception-test repairs of Section 5, the conformal stage is the only reason any entrant passes Kupiec at 97.5%. The guarantee also attaches to the shifted quantile alone, not to the deployed curve of (5), whose tail branch takes the more conservative of the shifted and EVT values and is then monotone-rearranged. The exact finite-sample coverage identity need not survive that composition, so the coverage claim at the reported levels rests on these exception tests rather than on Proposition 1. Conformal constructions with validity under dependence exist (Chernozhukov et al., 2018), but we prefer the simple split form plus an empirical audit.

One further implementation gap: the first-stage GARCH is fit on the full pre-test history, so the calibration errors are held out from the shape learner and the EVT tail but not from the filter. The accuracy-layer comparisons never read the calibration split and share their filter with the benchmark, so their information sets match. Only the overlay’s shift uses the split. Re-estimating with the filter stopped at the calibration boundary leaves the accuracy advantage intact (DM 5.1 and 5.4). The strict estimator’s gap to the full-window benchmark (-0.6 , -2.5) matches that benchmark’s own margin over its short-window twin (2.8 , 3.8), so the gap reflects a quarter less estimation data rather than leakage. The static shift widens (per-name pass 34% against 48%), consistent with its disclosed conservatism. Under the same strict split the adaptive conformal update of Gibbs and Candès (2021) walks the shift from -0.51 to -0.20 , lifts the per-name pass rate to 57%, and shrinks the joint-score gap to GARCH- t from DM -3.2 to -1.5 (not significant at 5%). The estimator’s annual-refit schedule sits between the static and adaptive poles, and the exception tests decide between them.

The full forecast composes the stages:

$$\widehat{Q}_t(\tau) = \widehat{\mu} + \widehat{\sigma}_t \cdot \mathbf{R} \left[\min\{\widetilde{q}_\phi(\tau \mid s_t), \widehat{q}^{\text{EVT}}(\tau)\} \mathbb{I}\{\tau \leq p_0\} + \widetilde{q}_\phi(\tau \mid s_t) \mathbb{I}\{\tau > p_0\} \right], \quad (5)$$

where $\mathbf{R}[\cdot]$ is the monotone rearrangement across levels. The conditional scale $\widehat{\sigma}_t$ carries the day's volatility from the GARCH stage, and the bracketed curve carries the shape from the nonparametric and EVT stages. A forecast is therefore the conditional mean plus scale times the modeled residual quantile: $\widehat{\text{VaR}}_{\tau,t} = \widehat{\mu} + \widehat{\sigma}_t \widehat{Q}_t^z(\tau)$ and $\widehat{\text{ES}}_{\tau,t} = \widehat{\mu} + \widehat{\sigma}_t \tau^{-1} \int_0^\tau \widehat{Q}_t^z(u) du$, with $\widehat{Q}_t^z$ the bracketed standardized-residual curve. The tail forecast is the pointwise more conservative branch (the body branch is the minimum on 38% of tail nodes), and both risk measures come from this one curve. Recomputing ES_α as the numerical integral of (5) rather than the GPD closed form leaves each comparison in place. At 2.5%, FZ0 is 1.849 against 1.851, the integral marginally better, DM 5.3, and the estimator-over-GARCH DM is 4.8 and 5.1 at the 1% and 2.5% levels. The splice level is immaterial ($p_0 \in \{1.5\%, 2.5\%, 5\%\}$: DM 4.6–4.8).

Industry models as special cases. Equation (5) nests the standard toolkit. Historical simulation sets $\widehat{\mu} \equiv 0$, $\widehat{\sigma}_t \equiv 1$, and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS) takes q as the unconditional empirical quantile of past *residuals*, i.e. Stage 2 with the state dependence deleted. GARCH- t takes $q = t_\nu^{-1}$, a fixed parametric shape. Whatever FHS captures, (3) can represent, and it adds conditional shape. CAViaR sits outside the nesting. It autoregresses on the quantile directly, with no scale/shape split, and is the benchmark with which the estimator ties (Section 5).

Algorithm 1 (residual-hybrid estimator). (i) Fit (2) per asset on a trailing window. Store $\hat{\sigma}_t, \hat{z}_t$. (ii) Pool $(s_{j,t}, \hat{z}_{j,t})$ across names. Train the shape model by (3). (iii) Fit the GPD tail (4) on past exceedances. Splice at p_0 . (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data. (v) Forecast by (5). In *production* all stages refit on an annual walk-forward schedule. The paper’s *evaluated* protocol freezes each stage once per split, the more conservative test since no stage ever adapts to the test era. Section 3.1 verifies the frontier separately under the true annual-refit schedule.

[Figure 1 about here.]

2.3 Estimating the shape: gradient boosting and neural quantile generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3). We use trees for point accuracy and the network when draws are needed (Online Appendix Table OA.3).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta h_m(\cdot)$ is grown stagewise. Each new tree h_m is fit to the negative gradient of the pinball loss, which equals τ_k where the current model under-predicts and $\tau_k - 1$ where it over-predicts. The fitted grid is made non-crossing by monotone rearrangement, which sorts the values $\{q(\tau_1|s), \dots, q(\tau_K|s)\}$ in τ and never increases the loss (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest point estimator, about 0.75% better pinball than the tuned network, consistent with the broader tabular-learning literature.

The neural implicit quantile network (IQN), the GBC estimator. A τ -conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s, \tau) = g(\psi(s) \circ \varphi(\tau)), \quad \varphi_j(\tau) = \text{ReLU}\left(\sum_{i=0}^{n_c-1} \cos(\pi i \tau) w_{ij} + b_j\right), \quad (6)$$

where $\circ$ is elementwise multiplication and g, ψ are feed-forward nets. Three modifications make it

competitive on financial data. The first is tail-aware τ -sampling, $\lambda = \frac{1}{2}U(0, 1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$, which oversamples the extremes that uniform sampling starves (the raw net’s 99% breach rate was 3.2%). The second is monotone rearrangement of the τ -curve, and the third is the conformal layer of Section 2.2. The network’s role is generation rather than point accuracy: with $\tau \sim U(0, 1)$, $z^* = H_\phi(s, \tau)$ draws from the fitted conditional law by inverse transform. The network learns one continuous state-conditional quantile map, so draws are available directly, without separately fitting and interpolating a grid of quantile models. A GARCH- t or FHS model also simulates, from its fixed innovation law. The trees need the interpolation step.

How training works. The network minimizes the empirical version of (3) by stochastic gradient descent, each example drawn at a fresh level $\tau_{jt} \sim \lambda$ each epoch, so one set of weights carries the entire quantile curve. Early stopping is on held-out pinball loss. Monotone rearrangement and the conformal shift are applied *after* training, and the whole pipeline is refit on the annual walk-forward schedule using only past data. Layer sizes, learning rates, and seeds are reported with the code release.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names. In generative-Bayes terms, H is an amortized map evaluated at any new (s, τ) without per-asset refitting. It works, at cold-start in particular, because the mapping is estimated across names rather than within one. A per-asset GARCH learns *name-specific parameters* from that name’s own history, so with no history it has nothing. The amortized model instead learns a single mapping from observable state to residual shape, estimated from millions of asset-days across names. A newly listed asset is therefore not a new problem: its first ticks and characteristics place it at a point of the state space already densely populated by other names’ histories, and the estimated mapping applies there directly. As the name’s own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically. For the same reason the explicit own-history

blends add nothing: the state vector already carries the name’s own recent information. Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name’s own recent dynamics, realized vol above all. Removing it costs +1.0%, while removing all cross-sectional characteristics costs only +0.5%. Characteristics matter at cold-start. Two explicit hierarchical corrections, a naive quantile blend toward own-history and an amortized-as-prior affine Gibbs update, both hurt at each listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly. Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 (benchmark tier) on the M5 competition data, outside finance altogether.

Estimator choice by forecasting objective. The two estimators minimize the same loss over different index sets. The trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\hat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level, while the network solves $\min_\phi \sum_i \mathbb{E}_{\tau \sim \lambda} \rho_\tau(\hat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum. The choice follows from what the application consumes. At fixed regulatory levels (VaR/ES reporting, limits, capital) the trees are preferable: best point accuracy, no GPU, no sampling. Where the application consumes draws (scenario generation, portfolio aggregation by simulation), the network is preferable. It learns one continuous conditional quantile map, so draws come by inverse transform with no grid interpolation, at a measured cost of $\sim 0.75\%$ in point accuracy. A GARCH- t or FHS model also simulates, from its fixed innovation law, whereas a tree grid requires monotone interpolation first. If both are needed, run both from the same pooled residual panel. They share each upstream and downstream stage of (5), so the swap is one component, not a new pipeline.

The amortized estimation loop is stated as Online Appendix Algorithm OA.1.

2.4 Formal results: why shape misspecification must show up in the loss

The paper’s organizing claim, that the nonparametric edge lives where the fixed innovation shape is locally wrong, is an empirical finding with a structural motivation in the pinball loss. The results below are deliberately modest: a shape violation creates a quantile wedge, and regret is locally quadratic in that wedge. They motivate looking where trailing shape diagnostics are extreme, but they do not prove that mk_{63} (Section 2.6) estimates the wedge. That link is established empirically.

Lemma 1 (pinball regret identity). *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q , finite by the moment condition. Then for any q ,*

$$S(q) - S(q_F) = \int_{q_F}^q (F(u) - \tau) du \geq 0, \quad (7)$$

with equality iff $F(u) = \tau$ for Lebesgue-almost each u between q_F and q . If F has density bounded as $0 < m \leq f \leq M$ on that closed interval,

$$\frac{m}{2} (q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2} (q - q_F)^2. \quad (8)$$

Proof sketch. The difference quotients of $q \mapsto \rho_\tau(Z - q)$ are dominated by the Lipschitz constant $\max(\tau, 1 - \tau)$, so differentiation under the expectation is justified and $S'(q) = F(q^-) - \tau$ (a.e. equal to $F(q) - \tau$). Integrating from q_F to q gives (7) for any F , atoms forming a null set. The integrand $F(u) - \tau$ has the sign of $u - q_F$ (by definition of q_F), so the integral is ≥ 0 on whichever side of q_F the point q lies. Inequality (8) then follows from $F(q_F) = \tau$ (supplied by the density hypothesis) and $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$ for almost each u between q_F and q . Full details, including both orientations of the integral, are in the Online Appendix. $\square$

Lemma 1 says the extra loss a misspecified forecaster pays is *locally quadratic in its quantile*

error. The frontier claim then reduces to whether shape misspecification creates a quantile wedge at the risk levels that matter. For the paper’s canonical case, a fixed Student- t shape when the true local shape is fatter, it must:

Proposition 2 (tail wedge under tail-shape misspecification). *Let $Q_\nu(\tau)$ denote the τ -quantile of the unit-variance Student- t with $\nu > 2$ degrees of freedom. For each pair $\nu_2 > \nu_1 > 2$ there exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2]$ such that $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$ for all $\tau < \tau^*$. The fatter-tailed law thus has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \rightarrow \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2} (Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$ at each level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof sketch. The unit-variance t_ν ($\nu > 2$, so the variance is finite and the normalization is well defined) has a regularly varying left tail: $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ as $x \rightarrow \infty$ with $c_\nu > 0$. Unit-variance rescaling alters c_ν but not the index ν . Inverting a regularly varying tail (Embrechts et al., 1997) gives $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1 + o(1))$ as $\tau \downarrow 0$. For $\nu_1 < \nu_2$ the exponent $1/\nu_1 > 1/\nu_2$ makes $|Q_{\nu_1}(\tau)|/|Q_{\nu_2}(\tau)| \rightarrow \infty$, so the strict ordering and the diverging wedge hold for all τ below some $\delta > 0$. Taking $\tau^* = \min(\delta, 1/2)$ places τ^* in $(0, 1/2]$ as claimed. The proposition asserts existence of such a τ^* , not the location of the finite-sample crossover. The latter is $\tau_c \approx 0.0179$ for the t_3 -vs-normal case of Remark 1. The regret bound is (8). Full details are in the Online Appendix. □

Remark 1 (the crossover level). Unit-variance fat-tailed laws put more mass near the center as well as in the far tail, so the quantile ordering of Proposition 2 reverses at moderate levels. Numerically, the unit-variance t_3 has a less extreme 5% quantile than the normal (-1.36 vs -1.64) but a more extreme 1% quantile (-2.62 vs -2.33). The sign flip occurs at a single crossover level $\tau_c \approx 0.0179$

(solving $Q_{t_3}(\tau) = \Phi^{-1}(\tau)$ numerically for the lower-tail root). The deepest deployable level $\tau = 0.01$ therefore sits below the crossover, where the fat tail is strictly more extreme. The level $\tau = 0.025$ sits just above it: at 2.5% the unit-variance t_3 quantile is -1.837 , still less extreme than the normal’s -1.960 . Two of the paper’s design choices are direct consequences. First, CRPS-style whole-distribution scoring can rank a fat-tail-aware model below a thin-tailed one even when the tail is the decision-relevant region (Section 2.1), because body and far tail disagree in sign. Second, the deepest deployable level straddles the crossover rather than sitting cleanly beyond it, and the score of Section 2.6 is used as a rank rather than a moment estimate.

The tail-wedge result of Proposition 2 motivates, but does not on its own establish, the empirical regularity that organizes the paper: that the nonparametric edge is ordered by the misspecification score. We do not claim a monotonicity theorem at the deployable levels, because two features of the standardized problem block one. First, Proposition 2 orders a given pair of tail indices only below a pair-specific level $\tau^*(\nu_1, \nu_2)$. It does not deliver monotonicity of the wedge in the tail index at a fixed operational τ . Second, the innovations are unit-variance by construction, and for unit-variance laws the level- τ quantile is not monotone in tail heaviness. As the tail fattens, the variance normalization pulls the fixed- τ quantile back toward the center (the crossover of Remark 1), so that at $\tau = 0.025$ a heavier-tailed t can carry the *less* extreme quantile. We therefore treat the score-ordering as an *empirical* regularity, documented below across the design panel, the frozen holdout, the FRTB battery, and the cross-market universes, and motivated by the tail-wedge mechanism rather than implied by it. The one exact statement we lean on is the regret identity (Lemma 1), the elementary basis for the consistency of the pinball loss (Gneiting, 2011; Ehm et al., 2016). Wherever the benchmark’s reported quantile differs from the oracle’s at a level, it pays a pinball regret that is positive and, to second order, quadratic in that gap. The score is our observable proxy for where that gap is largest.

Lemma 2 (validity of the generative sampler). *If $\tau \mapsto \tilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0, 1)$, then $Z^* = \tilde{q}(\tau \mid s)$ has quantile function exactly $\tilde{q}(\cdot \mid s)$.*

Proof sketch. For any x , left-continuity and monotonicity make $\{t \in (0, 1) : \tilde{q}(t \mid s) \leq x\}$ an interval closed at its top, so $\{\tilde{q}(\tau \mid s) \leq x\} = \{\tau \leq G(x)\}$ with $G(x) = \sup\{t : \tilde{q}(t \mid s) \leq x\}$. Since $\tau \sim U(0, 1)$, $\Pr(Z^* \leq x) = \Pr(\tau \leq G(x)) = G(x)$, i.e. Z^* has CDF G . G is the generalized inverse of $\tilde{q}$, and $\tilde{q}$ is nondecreasing and left-continuous. The generalized inverse of G therefore returns $\tilde{q}$ exactly (the standard inverse-of-inverse duality), so $\tilde{q}(\cdot \mid s)$ is the quantile function of Z^* . The full statement is in the Online Appendix. $\square$

Remark 2 (nesting and ERM dominance). The hypothesis class of Stage 2 contains the constant-in- s map $q(\tau)$, i.e., FHS. Empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Section 5 test whether that dominance survives estimation noise. It does. The same argument does not apply to CAViaR, so CAViaR is the one rival the estimator only ties.

2.5 Comparison with generative Bayesian computation

Relative to the generative Bayesian computation of Polson and Sokolov (2023), the downside-risk target changes five things in the estimator and one in the question. First, we train on a tail-weighted sampling measure $\lambda = \frac{1}{2}U(0, 1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$ rather than $\tau \sim U(0, 1)$, because uniform sampling starves the extreme levels downside risk depends on. Second, we model GARCH-standardized residuals rather than raw returns, keeping the conditional-scale dynamics the parametric model already estimates well. Third, we splice the GPD tail (4) at p_0 so that extrapolation beyond the training support is governed by extreme-value theory rather than the network. Fourth, the conformal stage adds an exchangeability-based coverage statement (Proposition 1). Fifth, the

amortization is different in kind: we amortize across the real cross-section of hundreds of assets rather than across simulations of a single model, and this produces the cold-start forecasts and transfer results of Section 2.3. Finally, the empirical question here, when such machinery improves on the parametric filter, is one generative computation does not address. The score below answers it.

2.6 The misspecification score

We use “misspecification score” as shorthand for a diagnostic of local residual-shape stress under the fitted GARCH- t benchmark. It is not a formal specification test.

GARCH- t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student- t with fixed shape. Static heavy tails do not violate this, since ν absorbs them. A locally wrong shape does: recent residuals far fatter-tailed or more asymmetric than any fixed t . The score is therefore computed per asset-day over the trailing window $W_t = \{t - 63, \dots, t - 1\}$ of $n_w = 63$ residuals, with window mean $\bar{z}$ and standard deviation s_z . The window *ends the day before the forecast target*, so each signal is lagged at least one day relative to the loss it is paired with. The frontier sorts of Section 4 pair S_{t-1} with the day- t loss, an alignment the estimation code enforces by construction. The two moment statistics are

$$\text{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\hat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \quad \text{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\hat{z}_i - \bar{z}}{s_z} \right)^3, \quad (9)$$

In the released code `mk63` and `sk63` are the bias-corrected sample excess kurtosis and skewness (pandas’ `kurt` and `skew`). At the full window length the correction is a common affine rescaling of (9), so the pooled decile ranks that the frontier uses are unchanged. *Skewness* is the signed third moment `sk63`. *Asymmetry* always means its absolute value $|\text{sk}_{63}|$, departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign. The score also

uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-5, \dots, t-1\}} |\widehat{z}_i|$. The frontier orders the edge by the decile of each of these signals. The deployed composite is the maximum of their trailing percentiles, $\text{score}_t = \max\{\widehat{F}_t(\text{mk}_{63}), \widehat{F}_t(|\text{sk}_{63}|), \widehat{F}_t(J_5)\}$, where $\widehat{F}_t$ is each signal’s percentile in the trailing panel available before t . The composite is itself deciled. All three are predetermined (trailing-window) and computable on any asset with a fitted GARCH, so the frontier is a deployable monitor rather than an ex-post classification. Online Appendix Table OA.3 reports the associated per-decile actions.

Why these statistics. Conditional on the fitted location-scale specification (μ_t, σ_t) , the distributional restriction the flexible family relaxes is the fixed t_ν innovation *shape*. The lowest-order ways a fixed symmetric law can be locally wrong are the third and fourth standardized moments. The statistic sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies. An exact t_ν sample has known positive excess kurtosis, but the score is used as a rank, so only its cross-sectional and temporal variation matters, not its t_ν baseline. J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent, and Section 4 shows the realized edge concentrates there. A signal that tracks departures from that fixed shape is a natural axis along which flexible-shape models can win.

The deployed monitor is stated as Online Appendix Algorithm OA.2.

3 Data and evaluation protocol

Equity panels come from CRSP (the Center for Research in Security Prices) daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations. Each modeling choice in the paper was made on this panel. The untouched-era holdout covers 2000–2013 and was pulled once, after the specification was frozen: the top 200 common shares (share codes 10 and 11) by average

market capitalization among those with at least 3,000 observations in the window. Returns are in percent and include CRSP delisting returns where present. The one delisting event in the design panel moves headline results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data. Terminal access ended in mid-2026, so those exhibits are frozen at the version reported here and all continuing work runs on WRDS alone.

Licensed rows never leave the machines they were pulled to. The public repository carries code, configuration, and derived statistics. The panels rebuild from the queries above for any licensed subscriber. Table 1 gives the sample flow.

[Table 1 about here.]

Splits are chronological within each name. The first 60% of a name’s history is the estimation window and the last 40% is test. The final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH- t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on pooled estimation-window residuals, the GPD tail on the same pooled residuals below their empirical 2.5% point, and the conformal shifts on the pooled calibration split. No stage refits inside the test window unless a rolling variant is named explicitly. Two structural properties of this protocol, calendar overlap in pooled training and the universe rule’s survivorship, are examined in their own subsections at the end of this section.

Loss and test conventions are collected in the note that closes this section. Distributional accuracy is scored by mean pinball loss across twelve quantile levels, and joint (VaR, ES) pairs by the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019). Loss comparisons use per-date Diebold–Mariano statistics with a Newey–West variance, and multi-model fields use the

Model Confidence Set. Exception behavior is tested by Kupiec, Christoffersen, and date-clustered breach tests at the two regulatory levels.

“Frozen-specification” has a specific meaning here: a run is frozen-specification when the full specification (signals, features, hyperparameters, split rule, and test statistics) is fixed and the predictions are written down before the data are pulled. We do not use the word *registered*, since no third-party registry or timestamp exists for these runs. The holdout pipeline is line-for-line the design-era pipeline. Both files are public in the replication repository, so the two can be compared directly to confirm that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 4 ran under this rule. As a post-hoc check, the deployed composite also replicates on this holdout. Its top decile carries +1.12% (DM 2.51), marginally above its kurtosis component (+1.04%, DM 2.16), and +0.71% (DM 2.08) under the expanding prior-only cutoff. Because the composite was not the rule frozen at the time, this is replication on an already-examined sample, not a frozen-specification test. Frozen-specification runs that remain open are listed in Section 7.

3.1 Common shocks, calendar overlap, and the pooled learner

Two nested flexible specifications appear in the paper. The deployed residual-hybrid holds the GARCH conditional scale fixed and models the standardized residuals; it produces the design-panel frontier (Section 4), the annual-refit check below, and every VaR and ES forecast. A simpler returns-space variant, which predicts return quantiles directly from lagged returns and trailing volatility and is plain enough to fix in advance, carries the pre-committed out-of-sample and robustness battery: the calendar-time split, the pre-2007 cut, the point-in-time universe, the untouched-era holdout, the family-wise multiplicity grid, and the timing and stability diagnostics. Both isolate where a flexible model beats the frozen parametric benchmark, and the residual-hybrid, holding

scale fixed, isolates innovation shape.

The design panels are split per name, 60/40 in each name’s own time index. In an unbalanced panel such a rule can place one name’s training window on another name’s test dates. A pooled learner could then meet a systemic event in training that a per-name model would only meet out of sample, a leak that needs no individual feature to look ahead. A per-asset split therefore cannot by itself support a causal reading of a pooled edge. Near-complete histories put the split points on almost the same date, so the channel here is nearly degenerate, and a strict calendar-time split closes it. Each stage (each GARCH fit, the pooled learner, the GPD tail) is re-estimated on data strictly before 2020-01-01 and tested from that date for each name simultaneously, so no training window shares a single calendar day with any test window. The frontier survives. The top decile carries +2.47% at per-date DM 6.22 over 1,258 dates (per-name split: +2.7%, DM 6.5), and the overall edge is +0.71% (DM 1.87). The bottom decile is elevated at +1.3%, because a 2020–22 test era leaves few calm cells. The ordering is intact, with the top decile roughly doubling each other cell. The concentration is not a calendar-overlap artifact. A second cut targets the harder era: inside the 2000–2013 holdout, everything is re-estimated strictly before 2007-01-01 and tested on 2007–2013, so the global financial crisis is entirely absent from training for each name. The pooled learner still wins overall (+0.29%, per-date DM 3.08 over 1,762 dates) and the within-sample decile profile keeps its shape (top three deciles +0.45–+1.04% against +0.06–+0.09% in the bottom three). The frozen-threshold top cell of Section 4 is too thin in that era to test alone (3.4% occupancy, +0.85%, DM 0.7). The holdout spans the 2008 crisis, which lay in no training window, so the significant edge there is not an artifact of a calm era.

A last construction tests the schedule the paper recommends: the evaluated protocol freezes each stage once per split, so the score could in principle measure parameter staleness rather than shape. The true annual walk-forward refits the per-name GARCH on expanding data at each

January-1 cutoff 2020–2024, rebuilds the residuals and score under the refit parameters, and retrains the pooled learner. Under it the top decile carries +2.09% at per-date DM 4.43, still large and significant, while the overall edge is no longer detectable (−0.18%, DM −0.52). Refitting removes the average gap but not the concentration, so stale benchmark parameters do not produce the conditional result.

3.2 Cross-sectional dependence, family-wise error, and the universe rule

Each headline test collapses the cross-section to one number per date before any variance is estimated (Section 4), so common-date dependence enters through the date series. Bootstrap procedures resample dates, not asset-days.

The frontier grid spans three signals and ten deciles, and per-cell p -values ignore the thirty hypotheses’ joint draw. We therefore report Romano–Wolf step-down family-wise-error control over the full 3×10 grid. The inputs are per-date mean edges per cell, a stationary bootstrap over dates that preserves cross-hypothesis dependence (mean block 10 days, $B = 2,000$, with block 20 as sensitivity), and one-sided step-down adjusted p -values. Nine of the thirty cells survive 5% family-wise control under both block lengths, and all nine are frontier cells. Three are the top deciles of the three signals ($\text{mk}_{63} \ t = 9.4$, $\text{skew}_{63} \ t = 9.2$, $\text{jump}_5 \ t = 5.1$, adjusted $p < 0.001$). The rest are the ninth deciles of mk_{63} (adjusted $p = 0.022$) and skew_{63} , skew_{63} ’s seventh, and jump_5 ’s U-shaped far cells. Each mid-grid cell with an unadjusted t near two (mk_{63} ’s sixth decile at $t = 2.34$, jump_5 ’s seventh at 2.08, skew_{63} ’s first at 2.01) loses significance under familywise control. The headline claims rest on the cells that survive it. Any cell that loses significance under the adjustment is reported as statistically indistinguishable from zero in the text wherever it is discussed. Across the paper’s other exercises the guard against selection is replication rather than a single joint correction. The frontier’s decile grid, the pre-committed returns-space specification, is

family-wise controlled, was specified before the run, replicates on the untouched 2000–2013 holdout as the frozen kurtosis component, and is recovered under a strictly non-anticipating decile cutoff. The deployed residual-hybrid reproduces the same, larger concentration on the design panel and under annual refit. The exploratory single-instrument universes, which are not jointly controlled with the equity grid, are labeled as such and carry their own 93-comparison adjustment.

The holdout’s observation filter conditions on names present for most of the era. The frontier comparison is relative (both models score the same names), so the filter tilts the universe toward survivors without by itself favoring either model in the ranking. Selection on information unavailable at the start of the era is the more serious issue, and a point-in-time re-selection addresses it directly. That construction fixes the universe at the top 200 names by market capitalization as of the first quarter of 2000 and applies no minimum-history filter. Each name is kept through its delisting date, with the CRSP delisting return appended as the final observation. On the 164 of 200 Q1-2000 names that clear the symmetric fitting floors (36 dropped, 79 delisting returns ridden to the end), the overall edge is +0.58% with per-date DM 8.96 over 2,692 dates. The frozen design-era top bucket carries +2.94% at DM 6.6 (5.9% occupancy). The bottom bucket is +0.44%, DM 5.4, so in this era even calm cells lean positive, at one-seventh the top bucket’s size. The edge is larger than in the survivor-tilted holdout, so the observation filter biased the reported edge downward rather than upward. The construction still requires the symmetric fitting floors, so the 36 names the estimators cannot fit do not enter, and the estimand is the 164 that clear them.

3.3 Reporting and inference conventions

The reporting and inference conventions used in the empirical sections that follow are fixed here. Edges are percentage reductions in pinball loss, so larger is better for the named model. A “+2.7% edge” means the model’s average loss is 2.7% lower than the benchmark’s. Ratios are model-over-

benchmark loss ratios, so below 1 favors the model and $1 - \text{ratio}$ is the edge ($0.973 =$ a 2.7% win, $1.043 =$ a 4.3% loss). DM is the Diebold–Mariano t -statistic on the loss difference. For the per-date DM, losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$. The orientation is stated at each use (benchmark minus engine in the engine’s headline comparisons, so positive favors the engine). The test is then run on the resulting $\sim 1,100$ -date series with a Newey–West (lag-10) variance, so the effective sample is the number of dates, not the number of asset-days. Reported p -values are one-sided in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%). A t -statistic above 2.6 is significant under either convention, and marginal cases are flagged where they occur. A tie means the per-date DM cannot reject zero at the two-sided 10% level. The point estimate is reported with the statistic so the reader can judge the width, and failure to reject is not treated as proof of equivalence. Where a genuine equivalence statement is possible we make it as one. For the CAViaR comparison the 90% confidence interval for the loss difference, $[-0.05\%, +0.11\%]$ of the benchmark loss, lies wholly inside a $\pm 0.25\%$ margin, which is a two-one-sided-tests conclusion rather than a failure to reject. The Model Confidence Set is computed on the same per-date loss matrix with the range statistic and a stationary bootstrap (mean block length 10 days, $B = 1,000$ replications in the main battery, $B = 800$ in the CAViaR extension), following Hansen et al. (2011). Coverage numbers are realized frequencies against a stated nominal (0.05 target: 0.056 is slight under-coverage of risk, 0.032 over-conservatism). For credible-interval coverage, closer to nominal is better in both directions. Correlations quoted for the frontier relate a universe’s residual-kurtosis level to the size of the nonparametric edge across that universe’s members. No other significance marks are used.

4 The misspecification frontier

4.1 The within-universe result

On 200 CRSP names across 1,108 test dates (≈ 221.6 k asset-days), we bucket the per-day pinball edge of the deployed residual-hybrid over GARCH- t into deciles of each misspecification signal. Both forecasters share the GARCH conditional scale, so the frontier isolates the value of the flexible innovation shape rather than any scale advantage. Sorting forecast performance by a predetermined state is a conditional predictive-ability comparison in the sense of Giacomini and White (2006). Table 2 shows the result that organizes the paper.

[Table 2 about here.]

The edge is a top-decile phenomenon. The top mk_{63} decile carries +2.98% at per-date DM 10.5, the overall edge is +0.35% at DM 5.35, and the lower deciles are statistically indistinguishable from zero. The edge is large and significant where the misspecification score is high, and statistically indistinguishable from zero where it is not.

The deployed composite. Table 2 sorts on each signal separately. The monitor the paper deploys is the composite score, the maximum across the three signals' trailing-panel percentiles (kurtosis, asymmetry, jump). Evaluated directly, its top decile is a 10% cell carrying +2.79% (DM 8.9), and the lower nine deciles sit between -0.3% and $+0.35\%$. The edge survives the strictly non-anticipating expanding cutoff at +2.72% (DM 9.0). The three components each carry a top-decile edge on their own: +2.98%, +2.78%, and +1.96% (kurtosis, asymmetry, jumps). The maximum of decile ranks instead flags the union of the three top deciles, a 19% cell at +1.72%, so we report the re-deciled composite.

The score's content could reflect model-relative misspecification, departure from each name's

own fitted t_{ν_i} , or the absolute level of recent tail activity. Converting each mk_{63} into a percentile of the simulated null of 63-window kurtosis under the name’s fitted t_{ν_i} leaves the frontier essentially unchanged (top decile +3.0%, DM 8.5, against +3.0%, DM 10.5 raw). The normalization is nearly rank-preserving, however (Spearman 0.97, 82% top-decile overlap), so it cannot by itself separate the two. A discriminating test instead favors the raw-tail-activity reading. In a within-date cross-sectional (Fama–MacBeth) regression of the per-observation edge on the raw score and the ν -relative percentile, the raw score carries the predictive content ($t = 7.8$), while the ν -relative percentile has significantly negative incremental content ($t = -5.2$). Double-sorting agrees. The top-raw-kurtosis-quintile edge falls from +1.9% for the most fat-tailed (low- ν) names to +1.5% for the thinnest-tailed, and observations high in the ν -relative score but not in raw kurtosis carry no edge (+0.3%, DM 0.5). The signal is thus the absolute level of recent tail activity rather than a per-name ν -normalized surprise. The signal still measures shape stress relative to the fitted model, since mk_{63} is computed on the standardized residuals the model treats as stationary t_{ν} . Elevated values flag states in which the realized residual shape is heavier or more asymmetric than the fitted law typically produces. A correctly specified low- ν t_{ν} can itself generate high realized kurtosis, so the score is a stress indicator rather than a formal specification test. The data reject the finer per-name normalization. A plausible reason is that a name’s fitted ν is estimated from the same residuals and a low value is itself a symptom of the tail activity that matters, so dividing it out removes part of the signal.

Timing diagnostics. The score is computed from residuals through day $t-1$ and paired with the day- t loss. Online Appendix Figure OA.1 probes that alignment. The deployed lag gives the largest top-decile edge in this returns-space diagnostic (+2.69%, DM 6.5), and a five-day-old score retains +1.67% (DM 6.4). A deliberately contaminated same-day window, the alignment a look-ahead artifact would favor, yields less than the deployed lag (+1.91%), and a one-day-ahead placebo gives

+1.92%. Same-day sorting does not manufacture the edge.

Non-anticipating decile boundaries. The sort forms deciles once over the pooled test panel, so top-decile membership draws on the full-sample distribution of mk_{63} even though the score itself is non-anticipating. Re-forming the 90th-percentile cutoff at each date from only mk_{63} observed on strictly earlier dates (250-day burn-in, the rule the deployed monitor uses) leaves 86% of membership unchanged. The top-bucket edge holds at +2.45% (DM 8.3), essentially the headline, so the pooled-panel cutoff does not produce the concentration. A within-date cross-sectional decile, which is not the deployed rule and forces 10% occupancy on each date including calm cross-sections with no misspecified name, dilutes the edge to +0.26% (DM 2.2). The frontier ranks the absolute level of residual misspecification, not within-day position.

An absorption test. If the score were merely an omitted variable, a parametric-adjacent fix should capture the edge without a pooled learner: per-name FHS with the residual window reweighted toward days whose mk_{63} resembles today’s (Gaussian kernel). On the full panel the reweighted FHS is significantly worse than plain FHS overall (per-date DM -11.3) and in the top decile (pinball 0.541 against 0.444), while the pooled model beats plain FHS there by 2.7% (DM 8.8). The deterioration is consistent with effective-sample collapse: sharp univariate conditioning shrinks a name’s usable history exactly when today is unusual. Pooling across names makes state-conditioning affordable, because the amortized model of Section 2.3 borrows the shape information from each name’s stressed days.

Robustness of the score. Online Appendix Table OA.2 sorts test asset-days independently into quintiles of mk_{63} and of trailing volatility (rv_{63}). High volatility without a shape break carries no edge (top-volatility column, bottom four kurtosis quintiles: -0.57% to $+0.10\%$), and a shape break

without an active regime carries almost none (top-kurtosis row, calmest quintile: -0.10% , DM 0.7). The product state carries $+2.33\%$ with DM 6.4. The two ingredients interact: the score flags a currently wrong parametric shape, and volatility scales the cost of the error. A superior-predictive-ability test (Hansen, 2005) on the per-date loss panel against the full desk field (GARCH- t , FHS, EWMA, HS) returns $T^{\text{SPA}} = 0$, consistent $p = 1.00$: no rival’s advantage is positive, and each is individually worse with $t \geq 5.8$. The score needs no recalibration. Holding the design-era top-decile threshold fixed, the top-decile edge is positive in each test year from 2020 to 2024 (between $+1.7\%$ and $+3.4\%$), even though the cross-sectional 90th percentile of mk_{63} drifts upward in 2023–24. The score is a slow state, so monitoring it costs little: a median name enters the top decile less than once a year (0.68 entries) and a median episode lasts 24 trading days.

The untouched-era holdout. Each design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze the entire pipeline (signals, features, hyperparameters, split rule, and test statistics) and wrote down three predictions in advance: a positive and significant top-decile edge with a bottom decile near zero, a small positive overall edge, and a threshold-shaped decile profile matching Table 2. We then pulled CRSP daily returns for 2000–2013, an era no prior run in this project had touched. The pre-specified kurtosis-component frontier, a simpler returns-space variant frozen in advance, replicates: the top-decile edge is $+0.89\%$ with per-date DM 2.16 ($p = 0.016$, 1,465 dates), deciles one through nine sit between -0.18% and $+0.28\%$, and the overall edge is $+0.02\%$. The deployed residual-hybrid composite replicates as well (Figure 2): its top decile carries $+1.12\%$ at per-date DM 2.51. This is the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. The decile boundaries above are within-sample ranks of the holdout test era, the standard conditional-sort convention. The partition rule never touches a forecast or a loss, but a desk could not have applied it on day one. A day-

one-applicable restatement freezes the design era’s numeric decile boundaries (the pooled design test sample’s mk_{63} edges, top threshold 13.67) and applies them to the holdout as constants. Membership at (i, t) then depends only on name i ’s own past 63 days and a number fixed before any holdout day was scored. The threshold shape survives the frozen rule: bins eight through ten carry +0.82%, +0.92%, and +1.02% against −0.3–0.0% below, and the top-bin point edge slightly exceeds the within-sample sort’s +0.89%. The cell thins to 3.3% occupancy in this lower-kurtosis era, however, and the per-date test loses power (DM 1.11, insensitive to Newey–West lags 5–44). A second real-time rule needs no design-era constant at all. An expanding pooled 90th-percentile threshold over strictly earlier dates (250-date burn-in) restores natural occupancy (9.5%) and recovers the sort’s result with significance (+0.99%, DM 2.15, insensitive to Newey–West lags 5–44). The bottom bucket is −0.18% ($t = -1.4$). The conditional sort, the frozen constant, and the recursive rule agree on the size of the top-cell edge, and the recursive rule is the version a desk could have run from day one. The frozen thresholds are also tested on the point-in-time universe of Section 3.2, where they carry +2.94% at DM 6.6.

[Figure 2 about here.]

The overall signal–edge correlation is low (0.05–0.10) because the relationship is a threshold non-linearity, not a linear one. The top-decile Diebold–Mariano statistics are the appropriate summary.

4.2 The frontier across universes

[Table 3 about here.]

Table 3 shows the same axis at work across markets. The only decisive standalone single-asset wins are the hyperinflation and capital-control currencies, the Argentine peso and the Egyptian pound, which are 12–14% better with significance. Their residual kurtosis (~ 2000 – 3300) sits two

orders of magnitude above anything GARCH- t was designed to absorb. The country gradient is mediated by residual kurtosis rather than by the development label: frontier indices (kurt ~ 10) sit between well-behaved developed indices (kurt ~ 5) and hyperinflation FX on the same frontier. The Korea-2026 negative control sharpens the mechanism: a price crash with circuit breakers is not residual misspecification. Korean residual kurtosis stays in the ordinary 4–13 range, and GARCH- t significantly wins all twenty-three assets even inside the crisis window. The frontier is about the residual shape, not about headline turbulence.

The three exploratory universes contain 93 instrument-level comparisons. Treated as one family under a Holm step-down at familywise 5% (Online Appendix Table OA.4), they leave eight numerical survivors. One of these (PJM power) is already reclassified by the audit below as a log-return artifact on near-zero prices, leaving seven: the three volatility indices, Baltic freight, wheat, USDEGP, and Kenya. Sri Lanka, Argentina, Pakistan, and Nigeria drop from the individually significant list. Instrument-level entries are therefore exploratory. The replication evidence this section rests on is pooled and structural: the crisis-tier FX comparison (DM 4.12, $p \approx 2 \times 10^{-5}$, surviving any adjustment) and the kurtosis–edge gradients across universes.

A frequency corollary. In the five daily crypto series examined here, GARCH wins throughout (zero nonparametric wins). At hourly frequency the picture inverts: the neural IQN beats the full benchmark ladder, including filtered historical simulation, by $\sim 0.6\%$ with per-date DM $p < 10^{-7}$. Sample density is itself a frontier dimension: more observations per unit of regime raise the resolvable residual-shape signal. A systematic intraday map of the frontier is specified in advance and blocked only on tick-data access.

4.3 Regime dependence and model selection

In the amortized panel the edge concentrates in turbulence. The GARCH-minus-GBM pinball gap grows monotonically from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent, roughly fourfold, and both nonparametric estimators beat GARCH in each quintile (overall: GARCH 0.6543, GBM 0.6447, IQN 0.6478).

Gating, running GARCH on calm days and switching to the nonparametric model when the score is high, does not improve on continuous use of the flexible model. A classifier gate trained on the residual-shape signals routes 90.5% of days to the nonparametric model and matches always-nonparametric exactly (pinball 0.3987, against 0.4000 for GARCH). Its routing also runs backwards across score deciles: nonparametric wins on high-misspecification days are rare but large, while a classifier targets frequency. A regression gate on the edge magnitude fixes the direction (routing flat ≈ 1.0 , predicted-edge correlation +0.16) and still converges to always-nonparametric. The per-day oracle gap (4.4%) is unreachable because per-day model advantage is dominated by which day the tail event lands, and that is unpredictable from prior-day state. In deployment the amortized setting uses the nonparametric model throughout, since it does not underperform on calm days. The residual-hybrid is the choice where a guaranteed $\geq$ GARCH floor is wanted, and the frontier score serves as a monitor (Section 6). Detection latency is small (median zero) and the edge is back-loaded within episodes, so the lag costs little even in the per-name setting.

5 VaR and Expected Shortfall forecast evaluation

5.1 Head-to-head accuracy and significance

Table 4 reports the full battery on 140 CRSP names (155k observations, 1,108 dates).

[Table 4 about here.]

The stress-era replication. The battery was re-run on the pre-committed 2000–2013 holdout panel. Splits follow Section 3, so each name’s test window runs from roughly mid-2008 through 2013, the crisis and its aftermath. The EVT-tailed engine passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t -statistics of 0.17 at 99% and 0.35 at 97.5%, consistent with nominal at both regulatory levels through the crisis window. The two models most common on desks do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH- t is also well calibrated in this era (94% pass rate), as the frontier predicts. Calibration parity is the norm, and the engine’s added value in any era is the accuracy edge where the misspecification score is high. The conformal layer, calibrated on pre-crisis data, tilts conservative out of era (breach 2.21% at 97.5%, $t = -1.09$, not significant). The safety margin widens when the calibration era is calmer than the test era, and the error runs in the direction regulators prefer.

Adding SAV-CAViaR, a model absent from banks’ standard toolkit but a leading semiparametric academic benchmark, produces a statistical tie: CAViaR 0.3461 vs hybrid 0.3460 (DM 0.4, $p = 0.34$), and the 90% Model Confidence Set contains exactly these two. The CAViaR study’s common sample differs slightly from Table 4’s, which is why the hybrid’s loss level is 0.3460 there against 0.3551 in the table. Rankings, not levels, are comparable across the two runs. The hybrid is co-best with CAViaR rather than uniquely dominant, and it significantly beats every model in banks’ standard toolkit. SAV-CAViaR models one quantile level per fit, produces no ES (the FRTB metric) without an ad-hoc stage, and needs per-name history, so it cannot price new listings. The hybrid produces the full curve and prices new listings. Pooled exception rejections are not counted against CAViaR. Pooled asset-day tests ignore cross-sectional dependence and overstate rejection for every model, the hybrid included, so the date-clustered restatement is the appropriate reading. Two textbook FHS variants, per-name train-constant and per-name rolling 500-day trailing,

were run on the identical panel. Both lose to the hybrid, by DM 5.2 and 5.7 (pinball 0.3562 and 0.3569 against 0.3551), and they bracket the pooled variant the battery uses. A neural IQN used as the residual model loses to trees in its untuned form (0.3650 vs 0.3551, DM 13.3) and has a thin tail (breach₉₉ 3.2%). The tail-aware, monotone, conformally recalibrated version narrows the gap (0.3638 vs 0.3632, DM 3.03) and repairs calibration (breach₉₉ 0.92%). Trees remain the strongest tabular estimator, and the tuned IQN is competitive. The ES columns are a diagnostic rather than a ranking. Under the exact tail integral, each fat-tailed entrant’s predicted ES exceeds the realized mean below its own VaR, by 9% (raw hybrid, rolling FHS) to 22% (pooled FHS). The comparator conditions on each model’s own breach set. The exact integral replaces an earlier three-node tail average that understated predicted ES by 2–6%. The joint FZ0 score below is the ES ranking used in this paper.

The joint (VaR, ES) score. Because ES is not elicitable alone but the pair (VaR, ES) is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the battery was also scored under the FZ0 joint score

$$L_{\text{FZ0},\alpha}(r, q, e) = -\frac{1}{\alpha e} \mathbb{I}\{r \leq q\}(q - r) + \frac{q}{e} + \log(-e) - 1, \quad e \leq q < 0, \quad (10)$$

the 0-homogeneous member of the class (Patton et al., 2019). Lower values are better. On the large-cap sandbox subset used to fix the score’s orientation, a Gaussian-innovation GARCH is significantly worst at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, Gaussian innovations are decisively inadequate. Among the fat-tailed entrants the FZ0 differences on large caps are small ($|\text{DM}| < 1.4$), the frontier’s own prediction that the shape edge is flat on well-behaved names. On a high-kurtosis subset the ordering favors the shape-aware forecaster, directionally but not significantly at that width. The pre-specified full-panel run, reported next,

settles the magnitudes on the engine’s own forecasts.

[Figure 3 about here.]

On the full 200-name panel (221.6k test asset-days), VaR and ES are read from the coherent min-envelope curve Q^* and per-date DM inference is used throughout. The engine’s *accuracy layer*, the EVT-tailed forecaster with no conformal shift, attains the lowest FZ0 at both regulatory levels. At 1% it beats GARCH- t (DM 4.8) and FHS (DM 4.9), the two standard benchmarks the accuracy claim rests on. A per-name score-driven one-factor GAS model estimated by FZ0 minimization in the style of Patton et al. (2019) loses by more (DM 9.6 at 1%, 6.2 at 2.5%). We read that comparison as indicative only. A few names’ GAS optimizers destabilize at 1% and inflate the gap, and a three-start re-estimate settles the stable 2.5% comparison near DM 5.9. At 2.5% the engine again wins over GARCH- t (DM 5.1). This ranking survives the annual walk-forward refit of Section 3.1: re-estimating every stage at each January-1 cutoff and re-scoring the 2020–2024 panel, the accuracy layer keeps the lowest FZ0 at both levels, beating GARCH- t and FHS by DM 3.6 and 2.7 at 1% and 3.1 and 2.6 at 2.5% (one-sided $p < 0.005$). Refitting narrows these margins but does not close them, unlike the average pinball edge, which closes under the same refit. The accuracy layer is thus one configuration that attains the lowest FZ0 at both levels and passes the aggregate date-clustered exception test at both levels. Per-name diagnostics show residual cross-sectional calibration heterogeneity (84% and 86% pass rates at 99%, Online Appendix Figure OA.2). The one failed test is the pooled 155k-observation Kupiec at 97.5%, which at this sample size rejects very small deviations. The frontier concentrates the joint-score edge as it does the pinball edge: the top misspecification decile carries DM 2.4 at the 1% level against DM 0.2 in the panel bulk. Adding the conformal shift at 97.5%, whose calibration split includes the 2020 crash, widens the band out of era (breach 1.6% against the 2.5% target) and gives back the FZ0 advantage there. The static-overlay variant loses to GARCH- t on the 2.5% joint score (DM -2.0). The concession is

largest where the bands are widest: in the top misspecification decile the static shift’s joint-score deficit reaches DM -7.1 . The coverage property costs sharpness under a strictly consistent score, and its error direction is conservative (wider bands, fewer breaches than nominal). A desk that values the conservative coverage margin keeps the shift. One scored on FZ0 efficiency runs the EVT tail alone, which passes the exception battery on its own (Online Appendix Figure OA.2). Most of the concession reflects a frozen margin. The adaptive form of the shift (Gibbs and Candès, 2021), updated daily from the panel breach frequency, shrinks the margin out of era (from -0.35 to -0.10 , breach $1.8\% \rightarrow 2.1\%$) and erases the overall concession. The adaptive variant ties GARCH- t on 2.5% FZ0 (DM -0.3). The top-decile cost survives either shift (-4.8 vs -6.0), the price of a margin where the bands are widest. The static shift also worsens per-name coverage (Kupiec_{97.5} pass rate 47%, against 72% unshifted and 69% adaptive), because over-coverage fails the test from the other side. Each “lowest FZ0” headline refers to the accuracy layer.

Direct ES calibration backtests. Exception counts test VaR, not ES. Two direct ES calibration backtests, the McNeil–Frey exceedance-residual test (McNeil and Frey, 2000) and the Acerbi–Székely Z_2 statistic (Acerbi and Székely, 2014), complement the jointly consistent FZ0 loss above. Nolde and Ziegel (2017) and Bayer and Dimitriadis (2022) draw the distinction between backtesting the calibration of an ES forecast and comparing forecasts through a joint score. Both are computed on the engine’s own min-envelope forecasts over the same 200-name panel, with a stationary block bootstrap over the 1,108 test dates (Online Appendix). The Gaussian filter is worst on each ES metric, and among the fat-tailed entrants no single model dominates each diagnostic. The engine has the Z_2 closest to zero at 1% and ties GARCH- t at 2.5%. The engine and GARCH- t are the two models that pass the 1% Kupiec test. On the McNeil–Frey residual, FHS, which targets the empirical tail directly, is closest to zero. The calibration tests show that the engine’s tail is well

calibrated. The comparative result is the FZ0 loss, where the edge over GARCH- t and FHS is significant at both levels.

5.2 The ten-day horizon: a direct multi-day extension

The multi-day results in this subsection come from a direct amortized model, a pooled quantile learner trained on h -day cumulative returns with GARCH-derived features. The residual-hybrid estimator of Algorithm 1 is a one-day architecture, and no ten-day claim in this paper attaches to the deployed engine. Training labels whose h -day window crosses the split are excluded, so no label contains test-era returns. At the FRTB ten-day liquidity horizon (CRSP 2014–2024) the direct model’s pinball edge over $\sqrt{h}$ -scaled GARCH grows from $\sim 0.4\%$ at $h = 1$ to $\sim 1.4\%$ at $h = 10$, and holds in the 2020–2022 stress window (hybrid $1.248 < \text{FHS } 1.2605 < \text{GARCH } 1.262$). The design’s h -curve (Online Appendix Figure OA.3) rises from 0.45% at $h=1$ to 2.1% at $h=20$ under the same purge.

Tail severity is more consequential. Ten-day ES is the same exact tail integral as in Section 5, with the same model-dependent own-tail caveat. Each ten-day tail over-states its own-tail comparator, GARCH $\sqrt{h}$ most (predicted -21.1 vs realized -15.4) and the direct model least (-20.1 vs -17.3), and the direct model has the best ten-day breach_{97.5} (2.79% vs GARCH 3.30% against the 2.5% target). Fat tails compound with horizon, and $\sqrt{h}$ Gaussian-style scaling compounds the shape error. The temporal split, however, places the whole test window in 2020–2024, and the pre-committed rerun on 2000–2013 shows that the ten-day result is era-dependent. Out of era the two methods tie on accuracy (direct edge $+0.49\%$, date-level DM 0.07 , NW lag 10), and the calibration verdict reverses. The $\sqrt{h}$ -scaled GARCH- t ES is almost exactly calibrated through the crisis window (predicted -16.6 vs realized -16.6 , numerical integral, purged), while the direct model’s tail under-states it (-15.9 vs -17.8). The pooled ten-day tail, trained on the pre-crisis

segment, did not extrapolate to 2008-scale ten-day losses. The ten-day tail-severity advantage is therefore a 2014–2024 statement, and a desk holding the engine at $h = 10$ through a 2008-type era should keep the parametric $\sqrt{h}$ number alongside as the conservative envelope. The frontier itself is confirmed on 2000–2013 (Section 4). The pre-specified horizon comparison thus ends with a mixed verdict.

5.3 Cross-asset evidence

Across the 43-instrument sweep the relationship is heterogeneous. The volatility indices (VIX, V2X, VVIX) and Baltic freight post the largest accuracy gains and survive the multiplicity adjustment (Online Appendix Table OA.4), but their calibration is open. VIX’s breach-independence is marginal (one-sided $p = 0.041$), and credit and Baltic freight fail it, so all are reported as accuracy edges rather than calibrated wins. An apparent electricity win reversed once the return transformation was corrected for near-zero prices. The GPD tail also hurts some volatility indices, motivating the selective-application rule of Section 2.2. Instrument-level results and diagnostics are collected in the Online Appendix.

The scale–shape decomposition. Against a Realized GARCH (Hansen et al., 2012) fitted on intraday data, the daily-data core concedes FZ0 at both levels (DM 6.2 and 4.5). Re-fitting the shape learner, EVT tail, and score on the realized residuals, however, leaves the score within noise. On large caps the frontier is therefore largely daily-scale misspecification, while elsewhere (day-one, illiquid, cross-asset) it is the only such diagnostic available (Online Appendix).

6 Extensions and implications

The full deployment pipeline is stated in the Online Appendix.

Ex ante use of the score. Because the frontier score is a trailing window on GARCH-standardized residuals, it is available *ex ante*: a desk computes today’s score from today’s data and acts on it for tomorrow’s risk number. The evidence does not support day-level model-switching. As Section 4.3 shows, the per-day oracle gap is unreachable (predicted-edge correlation +0.16), because which day a tail event lands is not forecastable from prior-day state. A learned gate collapses to always-nonparametric. The edge is negligible over the calm $\sim 90\%$ of asset-days and concentrated (+3.0% pinball, DM 10.5) in the top misspecification decile and the most turbulent volatility regimes (a $\sim$ fourfold gap, Section 4.3). On calm days the nonparametric/residual-hybrid family is statistically indistinguishable from the parametric benchmark (noninferior within the -0.25% margin where tested), and it has the advantage in stress. The score therefore serves as a monitor rather than a switch. The one setting in which the score does drive a switch is the per-name/standalone model, where nonparametric quantiles underperform on calm days. There the score gates, and its short detection latency (median zero, mean two to four days) costs little because the edge is back-loaded within a residual-shape stress episode.

Regulatory capital. The hybrid engine is an internal-models VaR/ES candidate. Its EVT-tailed accuracy layer attains the lowest joint FZ0 loss and passes the aggregate date-clustered exception test at both levels (per-name diagnostics show residual cross-sectional heterogeneity). This paper makes no capital-release claim: the predicted-vs-realized ES wedge conditions on each model’s own breach set and doubles under the exact tail integral, so no defensible arithmetic converts it to capital. The narrower implication is that the ES-based component of an IMA charge scales with the ES forecast. Actual FRTB capital, however, also involves liquidity-horizon adjustment, stressed-period ES on a reduced factor set, constrained aggregation, non-modellable risk factors, and desk-level eligibility through backtesting and P&L attribution. No one-for-one mapping from a

single-name ES forecast to desk capital exists. Establishing calibration differences requires a jointly valid standard, so the FZ0 comparison of Section 5 is the formal VaR/ES ranking. The ten-day rerun on 2000–2013 shows $\sqrt{h}$ -scaled GARCH almost exactly calibrated through the crisis era while the direct-learned tail under-states it: multi-day capital economics would be regime-dependent in the direction that matters most.

A misspecification monitor. The score is a live, per-asset monitor. An asset entering its top decile sits where the parametric model’s conditional loss runs $\sim 2\text{--}3\%$ higher. Because the input is a trailing window on GARCH residuals, surveillance is real-time and cheap.

Cold-start risk for new listings. The amortized model in its characteristics-only form is the only entrant in this paper that can produce a conditional risk forecast for an asset with no return history. An IPO or newly listed ETF gets day-one quantiles from its characteristics and its first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names. The day-one forecaster is the raw amortized model, and the residual-hybrid estimator of Section 2.2 attaches once a scale filter is estimable. The amortized forecast beats own-history benchmarks at each listing age, and the edge is largest when the name is young ($\sim 6\text{--}10\%$ of pinball in the first trading month, full-scale panel). Below the 750-observation fitting floor, a per-name GARCH is not reliably estimable in this panel. The ablation in Section 2.3 traces this as history accrues: characteristics carry the forecast at birth, and the name’s own realized volatility takes over as it ages.

7 Conclusion

A fixed-shape parametric filter is hard to beat where its shape assumption holds, and for most assets on most days it holds well. This paper locates the exceptions with an observable signal.

We identify a state, indexed by a real-time score of residual-shape stress, in which a distribution-free family has a large and significant advantage over the parametric benchmark. We build an engine that captures that advantage while matching the benchmark elsewhere. Judged by FRTB-aligned diagnostics, the engine improves on the market-risk models banks commonly run. The full internal-models-approval process is outside the paper’s scope.

These conclusions carry qualifications. The score is predictive rather than a necessary or sufficient condition for the flexible model to win. High residual kurtosis coincides with ties in some markets and with a nonparametric win at moderate kurtosis in others. In credit and freight, the accuracy edges sit alongside calibration failures that the current tail stages leave open. The score orders the edge as an empirical regularity, and the data do not identify residual misspecification as its structural cause. The evaluation is daily-frequency by design, and the one intraday comparison qualifies it. On large capitalizations where five-minute realized measures can be built, a realized-variance scale absorbs most of what the score detects on daily-GARCH residuals (Section 5). On that segment the frontier is largely a diagnostic of daily-scale misspecification rather than an independent conditional-shape phenomenon. Where intraday data do not exist, which is the estimator’s operating regime, the daily flexible model and its score remain the available tool. Finally, the score is a monitor rather than a rule for switching models from one day to the next. The day-ahead gap between the oracle and the achievable forecast is largely unforecastable noise (Section 4.3), so the score’s value lies in stratifying where the flexible model helps. Distributional uncertainty around these VaR and ES forecasts and the multivariate portfolio tail are left to a companion paper.

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Conflict of interest

The author declares no conflicts of interest.

Ethics statement

This study uses licensed secondary market data (CRSP/WRDS and Bloomberg) under the terms described in the data availability statement. It involves no human subjects and required no ethics approval.

Data and code availability

A public replication package covers every WRDS-based table and figure in this paper, with the Bloomberg-based exhibits preserved as-run from their derived series (terminal access having ended in mid-2026). It provides estimation code for all stages of the engine ((2)–(5)), the misspecification score (9), the FRTB battery, the generative-posterior and SBC routines, and the exact walk-forward, seeds, and hyperparameter grids referenced in the text. It is public at github.com/sean-qin-usa/downside-risk-paper (the repository README maps each table to its canonical script and result artifact. The strict-calibration audit is `code/job_fz_strict_calibration.py` $\rightarrow$ `results/fz_strict_calibration_results.json`). A

citable archival snapshot (Zenodo DOI) will be minted for the accepted version. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author’s institutional licenses and cannot be redistributed. The package includes the code to rebuild each panel from those sources, the derived non-proprietary series, and a synthetic example dataset (`toy_example.py`) that runs the full pipeline end-to-end without licensed data. The holdout’s frozen specification, including its written-in-advance predictions, is the header of `code/job_wrds_holdout.py`, line-for-line diffable against the design-era pipeline in the same folder. Two numerical corrections are canonical in the package: the GJR-skew- t inverse CDF (history preserved in `code/frtb_bench.py`) and the exact-tail integral ES. Table 4 is generated by `code/frtb_table.py` $\rightarrow$ `results/frtb_table_results.json`. The ten-day figures by `code/frtb_stress_exact.py` $\rightarrow$ `results/stress_es_results.json`. Pre-correction artifacts are preserved in the repository history, not in the working tree. The analyses deferred in earlier drafts and since completed ship with their result files, and the remaining open items (Section 7) are tracked in the repository.

Notes

¹The peaks-over-threshold formulation (Balkema and de Haan, 1974; Pickands, 1975) uses tail data more efficiently than block maxima and is standard in risk management (McNeil and Frey, 2000; Embrechts et al., 1997). Here $\xi > 0$ is the heavy-tailed case.

References

- Acerbi, C. and Székely, B. (2014). Back-testing expected shortfall. *Risk*, 27(11):76–81.
- Balkema, A. A. and de Haan, L. (1974). Residual life time at great age. *Annals of Probability*, 2(5):792–804.

- Barone-Adesi, G., Giannopoulos, K., and Vosper, L. (1999). VaR without correlations for portfolios of derivative securities. *Journal of Futures Markets*, 19(5):583–602.
- Basel Committee on Banking Supervision (2019). Minimum capital requirements for market risk. Technical report, Bank for International Settlements.
- Bayer, S. and Dimitriadis, T. (2022). Regression-based expected shortfall backtesting. *Journal of Financial Econometrics*, 20(3):437–471.
- Bissiri, P. G., Holmes, C. C., and Walker, S. G. (2016). A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.
- Chernozhukov, V., Fernández-Val, I., and Galichon, A. (2010). Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125.
- Chernozhukov, V., Wüthrich, K., and Zhu, Y. (2018). Exact and robust conformal inference methods for predictive machine learning with dependent data. In *Proceedings of the 31st Conference on Learning Theory*, volume 75 of *Proceedings of Machine Learning Research*, pages 732–749.
- Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4):841–862.
- Dabney, W., Ostrovski, G., Silver, D., and Munos, R. (2018). Implicit quantile networks for distributional reinforcement learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1096–1105.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

- Ehm, W., Gneiting, T., Jordan, A., and Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 78(3):505–562.
- Embrechts, P., Klüppelberg, C., and Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer.
- Engle, R. F. and Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4):367–381.
- Fissler, T. and Ziegel, J. F. (2016). Higher order elicibility and Osband’s principle. *Annals of Statistics*, 44(4):1680–1707.
- Giacomini, R. and White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578.
- Gibbs, I. and Candès, E. (2021). Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems 34 (NeurIPS)*, pages 1660–1672.
- Glosten, L. R., Jagannathan, R., and Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5):1779–1801.
- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494):746–762.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380.

- Hansen, P. R., Huang, Z., and Shek, H. H. (2012). Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6):877–906.
- Hansen, P. R., Lunde, A., and Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2):453–497.
- Jiang, W. and Tanner, M. A. (2008). Gibbs posterior for variable selection in high-dimensional classification and data mining. *Annals of Statistics*, 36(5):2207–2231.
- Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2):73–84.
- McNeil, A. J. and Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4):271–300.
- Nareklshvili, M., Polson, N. G., and Sokolov, V. (2025). Generative quantile Bayesian prediction. *arXiv preprint arXiv:2510.21784*.
- Nolde, N. and Ziegel, J. F. (2017). Elicitability and backtesting: Perspectives for banking regulation. *Annals of Applied Statistics*, 11(4):1833–1874.
- Patton, A. J., Ziegel, J. F., and Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and Value-at-Risk). *Journal of Econometrics*, 211(2):388–413.
- Pickands, J. (1975). Statistical inference using extreme order statistics. *Annals of Statistics*, 3(1):119–131.
- Polson, N. G. and Sokolov, V. (2023). Generative AI for Bayesian computation. *arXiv preprint arXiv:2305.14972*.

- Romano, Y., Patterson, E., and Candès, E. J. (2019). Conformalized quantile regression. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pages 3538–3548.
- Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.
- Zhang, C., Zhang, Y., Cucuringu, M., and Qian, Z. (2024). Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, 22(2):492–530.
- Zumbach, G. O. (2007). The RiskMetrics 2006 methodology. Technical report, RiskMetrics Group.

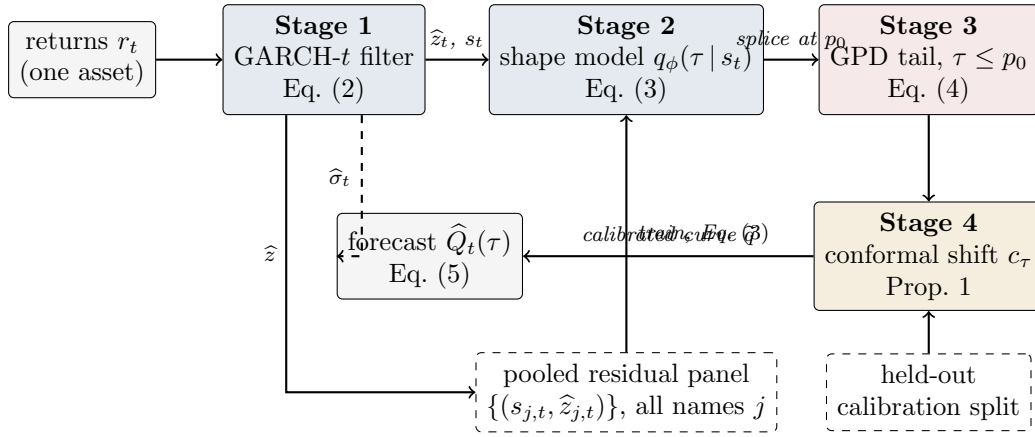


Figure 1: The residual-hybrid estimator as a pipeline. Solid arrows carry data. The dashed top arrow carries the conditional scale $\hat{\sigma}_t$ that rescales the finished residual quantile curve. Dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal overlay). In production each stage refits annually using only past data. The evaluated protocol freezes each stage once per split, and the annual-refit rerun is reported separately.

Alt text: Flow diagram of the residual-hybrid forecasting pipeline: daily returns feed a GARCH conditional-scale filter. The standardized residuals feed a state-conditioned nonparametric quantile model. An extreme-value left tail and an optional conformal overlay produce the VaR and ES forecasts.

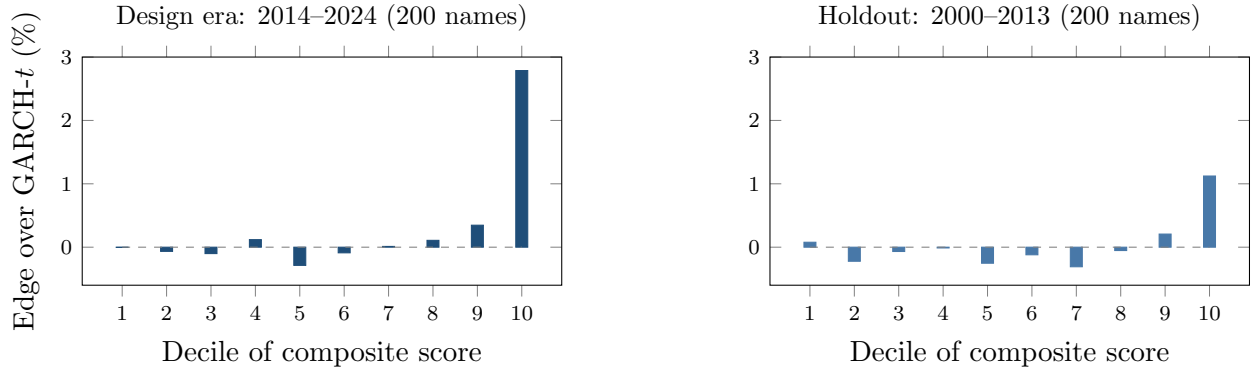


Figure 2: The frontier in and out of era, by decile of the composite misspecification score. Left: the design era (2014–2024). The top decile carries +2.79% at per-date DM 8.9, while deciles one through nine sit within $\pm 0.35\%$. Right: the deployed composite run once on 2000–2013 CRSP data. The top decile carries +1.12% at per-date DM 2.51 over 1,465 dates. The threshold shape (edge concentrated in the top decile, near zero elsewhere) survives an untouched era containing the 2008 crisis.

Alt text: Two panels plotting the nonparametric pinball-loss edge over GARCH- t by decile of the misspecification score. The edge is flat and near zero across the lower deciles and rises sharply in the top decile, in both the design-era panel and the 2000–2013 holdout.

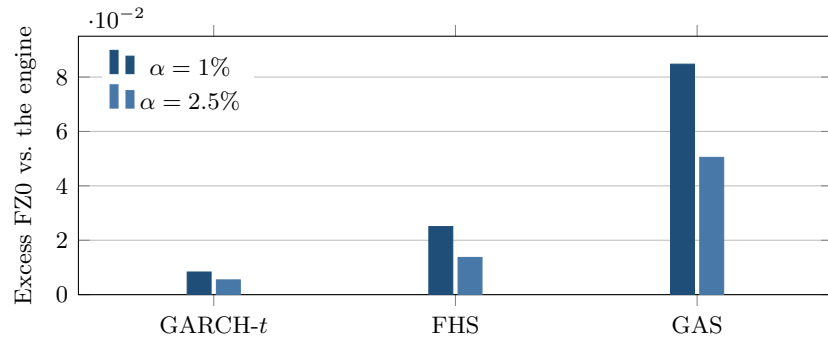


Figure 3: Joint (VaR, ES) FZ0 loss of each benchmark in excess of the engine’s accuracy layer, on the full 200-name panel. A positive bar means worse than the engine. The engine (the zero baseline) attains the lowest FZ0 at both regulatory levels, beating GARCH- t (DM 4.8 at 1%, 5.1 at 2.5%) and FHS (DM 4.9).

Table 1: Sample flow across evaluation panels.

Panel	Source	Era	Names	Test obs	Role
Design equity panel	CRSP/WRDS	2014–2024	200	221,600	frontier map, engine development
Untouched-era holdout	CRSP/WRDS	2000–2013	200	280,608	frozen-spec replication
Exception restatement	CRSP/WRDS	2014–2024	140	$\approx 155,000$	per-asset backtesting battery
Cross-asset sweep	Bloomberg	through 2026	43		^a frontier breadth
Country indices	Bloomberg	through 2026	26		^a frontier breadth

^a Frozen exhibits. Per-instrument windows vary and are documented instrument-by-instrument in the repository.

Table 2: The misspecification frontier: nonparametric edge over GARCH- t by decile of the trailing misspecification score (200 CRSP names, 1,108 dates, 221.6k asset-days). Edge = (GARCH pinball – nonparam pinball)/GARCH pinball.

Decile	Edge by signal decile (%)		
	mk ₆₃ (kurtosis)	sk ₆₃ (asymmetry)	jump ₅
1–8 (bulk)	≈ 0	≈ 0	mixed
9	+0.46	+0.59	—
10 (top)	+2.98	+2.78	+1.96
Overall	+0.35		

Per-date Diebold–Mariano: top mk₆₃ decile +2.98%, DM 10.5; |sk₆₃| +2.78%, DM 9.85; jump₅ +1.96%, DM 7.63 (all $p \approx 0$). Overall +0.35%, DM 5.35; the composite’s bottom decile is indistinguishable from zero (–0.00%, DM 0.36). The deployed composite monitor’s top decile carries +2.79%, DM 8.91 (Figure 2). Top-decile signal means: kurtosis ≈ 25 , |skew| ≈ 4 . jump₅ is U-shaped, its lowest decile also carrying an edge (+1.17%, DM 5.5), so its bulk is mixed rather than near zero.

Table 3: The frontier across universes. Ratio = nonparam/GARCH- t pinball (< 1 : non-parametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM -13.7
	EM (10)	1.013	8	GARCH, pooled DM -9.5
	crisis (7)	0.973	~ 1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86 , $p = .002$
	USDEGP	0.857	1920	DM 3.80 , $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins ^a
Cross-asset (43)	vol indices	~ 0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
	13/43 significant wins overall, $\text{corr}(\log \text{ kurt}, \text{ edge}) = 0.43$			
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM -2.5 to -5

^a Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06). Exploratory: only Kenya survives the 93-comparison familywise adjustment (Online Appendix Table OA.4). Cross-index $\text{corr}(\log \text{ residual kurtosis}, \text{ edge}) = 0.53$. Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating that kurtosis is neither necessary nor sufficient.

Table 4: FRTB battery: average pinball loss (twelve quantile levels), ES_{97.5} calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	ES _{97.5} pred. vs realized	Breach ₉₉	Kupiec ₉₉ p
<i>Ideal value</i>	<i>(smaller better)</i>	<i>(pred. = realized)</i>	<i>1.00%</i>	<i>> 0.05 (pass)</i>
Residual-hybrid (GBM)	0.3551	−6.37 / −5.83	1.09%	0.00
+ EVT tail	0.3555	−6.92 / −5.91	0.91%	0.00
GARCH(1,1)- t	0.3561	−6.69 / −5.65	1.06%	0.026
GJR-GARCH-skew- t	0.3562	−6.58 / −5.84	1.05%	0.062
GARCH-FHS (per-name)	0.3562	−6.61 / −5.84	1.11%	0.00
GARCH-FHS (pooled)	0.3566	−6.90 / −5.67	0.99%	0.79
GARCH-FHS (rolling 500d)	0.3569	−6.53 / −5.98	1.09%	0.00
EWMA (RiskMetrics)	0.3606	−5.52 / −5.63	1.63%	0.00
Historical simulation	0.3619	−7.21 / −6.86	0.84%	0.00

All columns are generated from a single common-sample run (`frtb_table_results.json`): the 140 CRSP names with the most complete history (at least 1,500 return observations each), over the common 1,108 test dates (155k asset-days), so each per-name exception test is adequately powered. The rolling-FHS burn-in then trims the same early dates from all models identically. Predicted ES is the numerical tail integral $\alpha^{-1} \int_0^\alpha Q(u) du$: closed forms for the t and normal entrants, 200-node integration of the Hansen skew- t inverse CDF, and a matched 20-node midpoint rule for the empirical quantile models and for the hybrid engine, whose VaR and ES are both read from one monotonized min-envelope curve $Q^* = \min\{\text{body}, \text{EVT}\}$. An earlier draft approximated predicted ES by a three-node tail-quantile average, understating it by 2–6%. The numerical integral is why these levels differ from prior versions. Diebold–Mariano vs the hybrid: GARCH- t 5.0, GJR-skew- t 5.0, FHS pooled 6.2, per-name 5.2, rolling 5.7, HS 7.5, EWMA 8.7 (all $p \approx 0$, rounded, and stable across reruns). The EVT variant concedes 0.0004 pinball to the raw hybrid (DM ≈ 6) as the price of its tail. The 90% MCS contains the raw hybrid alone, and CAViaR (separate common-sample run) statistically ties it.

The ES columns are a diagnostic, not a ranking: “realized” is the mean return below each model’s own VaR, so the conditioning set is model-dependent, and under the numerical integral each fat-tailed entrant exceeds its own-tail comparator by 9–22%. The joint Fissler–Ziegel score of the FZ0 re-scoring is the ranking criterion for ES.

GJR-skew- t row: corrected Hansen inverse CDF with the fitted η, λ (median 4.2, -0.015 , validated against the symmetric- t limit and `arch`’s own distribution), the original implementation silently reduced to a symmetric $t(8)$. Defect and correction are disclosed in the availability section.

Exception columns are pooled panel diagnostics: at 155k observations a breach-rate deviation of ± 0.1 points rejects in either direction (the EVT variant’s 0.91% sits on the conservative side, pooled FHS happens to land at 0.99%), so pooled p -values are reported, not leaned on. The regulatory reading is the per-asset and date-clustered restatement (Online Appendix Figure OA.2): per asset the EVT-tailed engine passes Kupiec at 99% for 84% of names and Christoffersen for 86%, and the date-clustered test passes at both levels ($t = 1.01$ at 99%, 1.86 at 97.5%) while the raw hybrid is rejected ($t = 4.5$): the tail stages carry the regulatory calibration, and the 2008-era battery re-run confirms it out of era. At 97.5% the split-conformal recalibration is the stage that repairs pooled Kupiec (GBM-recal $p = 0.24$).